

*Teaching Notes 7B**

Beyond MM: Bankruptcy

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*Many thanks to Prof. Amit Seru who made his material available to me. These teaching notes closely follow his.

Overview

- Focus on third MM assumption
 - Bankruptcy is not costly
- Allowing for costs associated with bankruptcy may impact capital structure
 - Costs called “cost of financial distress” or CFD
- Leads to “Trade-Off” theory of capital structure

Objectives

- Understand the “qualitative” aspects that are introduced when we relax MM assumptions
- Understand “quantitatively” what relaxing a particular assumption implies for capital structure

Assumption #3

Bankruptcy is not Costly

MM Assumptions: Review

- MM
 - Capital structure irrelevant for value of the firm
- But it only holds under certain assumptions:
 - Perfect and complete capital markets
 - No Taxes
 - Bankruptcy is not costly
 - Capital Structure does not affect investment policy and cash-flows
 - Symmetric information
- Now, relax third assumption
 - Bankruptcy is not costly

Economic versus Financial Distress

- When analyzing costs of bankruptcy, separate:
 - “*Economic Distress*” vs. “*Financial Distress*”
- **Economic Distress**
 - Operations of distressed firm performing poorly
 - Sales down, profits down, customers leaving, etc.
- **Financial Distress**
 - High leverage and presence of creditors cause further costs (CFD) and reduce firm value
 - Creditors demand concessions, difficulty accessing credit market, suppliers demand discounts

Example:

Understanding Economic Distress

Example: Understanding Two Types of Distress

- Armin's revenues have declined so it is going to introduce a new product
 - If product is a "Hit," Armin worth \$150m
 - If product "Flops," Armin worth \$80m
- 50-50 chance of "hit" or "flop"
 - Assume no discounting
- Suppose we are in a MM world
- Should we use equity or debt financing?
 - Debt due next year at \$100m
 - Note: \$100m is called "face value" of debt

Example (contd.): The “Hit” Scenario

- New project is worth \$150m if it “hits”
- With pure equity financing, $E = \$150$
- With debt financing, must repay \$100m to debt holders
 - So $E = \$50\text{m}$

Example (contd.): The “Flop” Scenario

- Armin Inc. is only worth \$80m if it “flops”
- With pure equity, $E = \$80\text{m}$
 - This is economic distress
- With debt financing, must pay \$80m to debt holders (we are in default)
 - This is both economic and financial distress
 - Debt holders lose \$20m
 - Equity is wiped out ($E = \$0$)

Example (contd.): Cost of financial distress?

Let's compare the two financing forms

- Equity financing
 - “Hit”: $E = \$150\text{m}$
 - “Flop”: $E = \$80\text{m}$
- Debt financing
 - “Hit”: $(E = \$50\text{m}) + (D = \$100\text{m}) = \$150\text{m}$
 - “Flop”: $(E = \$0) + (D = \$80\text{m}) = \$80\text{m}$
- Value to investors same ($V = \$115\text{m}$)
 - There is no cost of financial distress
 - Only economic distress matters in a MM world

Are there Costs of Financial Distress (CFD)?

- Chapter 7: Liquidation
 - Trustee appointed to sell / liquidate assets
 - Money distributed according to *strict priority*
 - Secured claims
 - Wages
 - Taxes
 - General unsecured claims (in order of seniority)
 - Equity
- Chapter 11: Reorganization
 - Debtor must present reorganization plan in court
 - If plan accepted, debtor keeps possession of assets, keeps running the firm.
 - Debtor gets automatic stay as protection against creditors (to prevent a run on the assets)

The 20 Largest Bankruptcies, 1980-2017

Company	Bankruptcy Date	Total Assets (pre-bankruptcy) in USD Millions
Lehman Brothers Holdings, Inc	9/15/2008	691,063
Washington Mutual, Inc.	9/26/2008	327,913
Worldcom, Inc.	7/21/2002	103,914
General Motors Corporation	6/1/2009	91,047
CIT Group Inc.	11/1/2009	80,448
Enron Corp.	12/2/2001	65,503
Conseco, Inc.	12/17/2002	61,392
Energy Future Holdings Corp.	4/29/2014	40,970
MF Global Holdings Ltd.	10/31/2011	40,541
Chrysler LLC	4/30/2009	39,940
Thornburg Mortgage, Inc.	5/1/2009	36,521
Pacific Gas and Electric Co.	4/6/2001	36,152
Texaco, Inc.	4/12/1987	34,940
Financial Corp. of America	9/9/1988	33,864
Refco Inc.	10/17/2005	33,333
IndyMac Bancorp, Inc.	7/31/2008	32,734
Global Crossing Ltd.	1/28/2002	30,185
Bank of New England Corp.	1/7/1991	29,773
General Growth Properties, Inc.	4/16/2009	29,773
Lyondell Chemical Company	1/6/2009	27,392

Two kinds of CFD: Direct and Indirect

- **Direct CFD:** a result of the default
 - Legal and administrative costs, payments to lawyers and accountants, restructuring costs
 - Cost of “fire sales”
 - Breaking up intangible capital such as human capital or brands
- **Indirect CFD:** due to “potential future” distress
 - Customers lost due to potential future distress (i.e., as a result of concerns about warranties, etc.)
 - Deterioration of relationships with suppliers and employees due to potential trouble
 - Agency problems that distort firm’s investments

How Large is Direct CFD?

- Old railroad study: Warner (1977)
 - Direct costs average 5.3% of market value
 - But < 1% compared to value 7 years earlier
- Broader study of NYSE/AMEX firms: Weiss (1990)
 - Direct bankruptcy costs = 3.1% of assets
 - Probability of bankruptcy = 0.7% per year
- Direct costs of bankruptcy are small
 - If cost a perpetuity, $PV(\text{Direct costs}) \sim V * 4\% * 1\% / 10\% = 0.4\% * V$
- $PV(\text{Direct CFD})$ tiny relative to $PV(\text{DTS}) = D * t$
 - If bankruptcy matters, must be because of indirect CFD

How Large is Indirect CFD?

- Potential Indirect Costs:
 - **Loss of Customers**
 - Hesitant to buy product if uncertainty about warranties
 - **Loss of Suppliers**
 - Hesitant to enter into long-term relationship/provide trade credit
 - **Loss of Employees**
 - Concerns about future employment (and pension benefits)
 - **Delayed Liquidation**
 - **Strategic Weakness**
 - ***Agency Conflicts***

How Large is Indirect CFD?

- Indirect costs are difficult to quantify
 - Two broad steps
- Step 1: Identify losses to unlevered firm value
 - Incurred on ongoing basis, even if company is not currently distressed
- Step 2: Identify incremental losses beyond losses arising from economic distress
- One estimate of indirect CFD is 10-20% of firm value: Andrade and Kaplan (1998), Veronesi and Zingales (2010, in banking sector).

Example:

Understanding Financial Distress

Example: What is CFD and Who Bears it?

- Remember the Armin example. We invest in a new product
 - A “Hit” is worth \$150m and a “Flop” is worth \$80m
 - We have to repay debt of \$100m
- Suppose we now relax the third assumption
 - Bankruptcy is costly
- Let’s say the cost of financial distress is \$20m
 - We can only repay \$60m to debt if we “Flop”
 - Expected $PV[CFD] = \$10m$

Example: What is CFD and Who Bears it?

Let's compare the two financing forms

- Equity financing
 - “Hit”: $E = \$150\text{m}$
 - “Flop”: $E = \$80\text{m}$
 - Value = $(1/2 * 150 + 1/2 * 80) = \underline{115}$
- Debt financing
 - “Hit”: $(E = \$50\text{m}) + (D = \$100\text{m}) = \$150\text{m}$
 - “Flop”: $(E = \$0) + (D = \$60\text{m}) = \$60\text{m}$
 - Value = $(1/2 * 150 + 1/2 * 60) = \underline{105}$
- Value with debt financing is lower due to CFD
 - But who pays the \$10m?

Example: What is CFD and Who Bears it?

- To understand, modify the Armin example
 - Need to know what the investment is and who funds it
 - Suppose the investment in the Armin project is \$80m
- Consider 2 scenarios:
 - Debt Financing: Face Value = \$80m
 - Debt Financing: Face Value = \$100m
- CFD is \$20m
- Let us examine how much the equity and debt holders make in the 2 scenarios
 - Similar to Two steps outlined before: In one scenario, the firm will incur CFD, while in the other scenario it will not

Example: What is CFD and Who Bears it?

- Scenario 1: Debt with Face Value = \$80m
 - “Hit”: $D = \$80m$; $E = \$70m$
 - “Flop”: $D = \$80m$; $E = 0$
 - Value of Equity = $(1/2 * 70) = \$35m$
 - Value of Debt = $(1/2 * 80 + 1/2 * 80) = \$80m$
 - Value = Value of Debt + Value of Equity = \$115m
- Debt sufficient to finance investment (\$80m) with \$0 CFD
 - Debt holders receive amounts in each state such that, in expectation at $t=0$, they recover exactly what they lend to the firm (\$80m)
- Value to Equity holders = \$35m

Example: What is CFD and Who Bears it?

- Scenario 2: Debt with Face Value = \$100m
 - “Hit”: $D = \$100m$; $E = \$50m$
 - “Flop”: $D = \$60m$; $E = 0$ [Note: \$20m lost due to CFD]
 - Value of Equity = $(1/2 * 50) = \$25m$
 - Value of Debt = $(1/2 * 100 + 1/2 * 60) = \$80m$
 - Value = Value of Debt + Value of Equity = \$105m
- Debt sufficient to finance investment (\$80m) with \$20m CFD
 - Debt holders (again!) make amounts in each state such that, in expectation at $t=0$, they recover exactly what they lend to the firm (\$80m)
- Value to Equity holders = \$25m
 - ➔ Equity holders “ex-ante” pay for CFD!

Combining Debt & Taxes with Costly
Bankruptcy:
Trade-Off Theory of Capital Structure

Trade-Off Theory of Capital Structure

- Remember: Goal is to think about implication of relaxing assumptions on capital structure
 - From perspective of decision maker in the firm
- Combine the effect of taxes and bankruptcy into a theory of capital structure
 - Leverage increases the $PV(DTS)$
 - But leverage also reduces value via $PV(CFD)$
 - Both the benefit of DTS and the CFD are borne by existing shareholders
 - Hence, shareholders/managers would prefer if we optimize by balancing the two effects

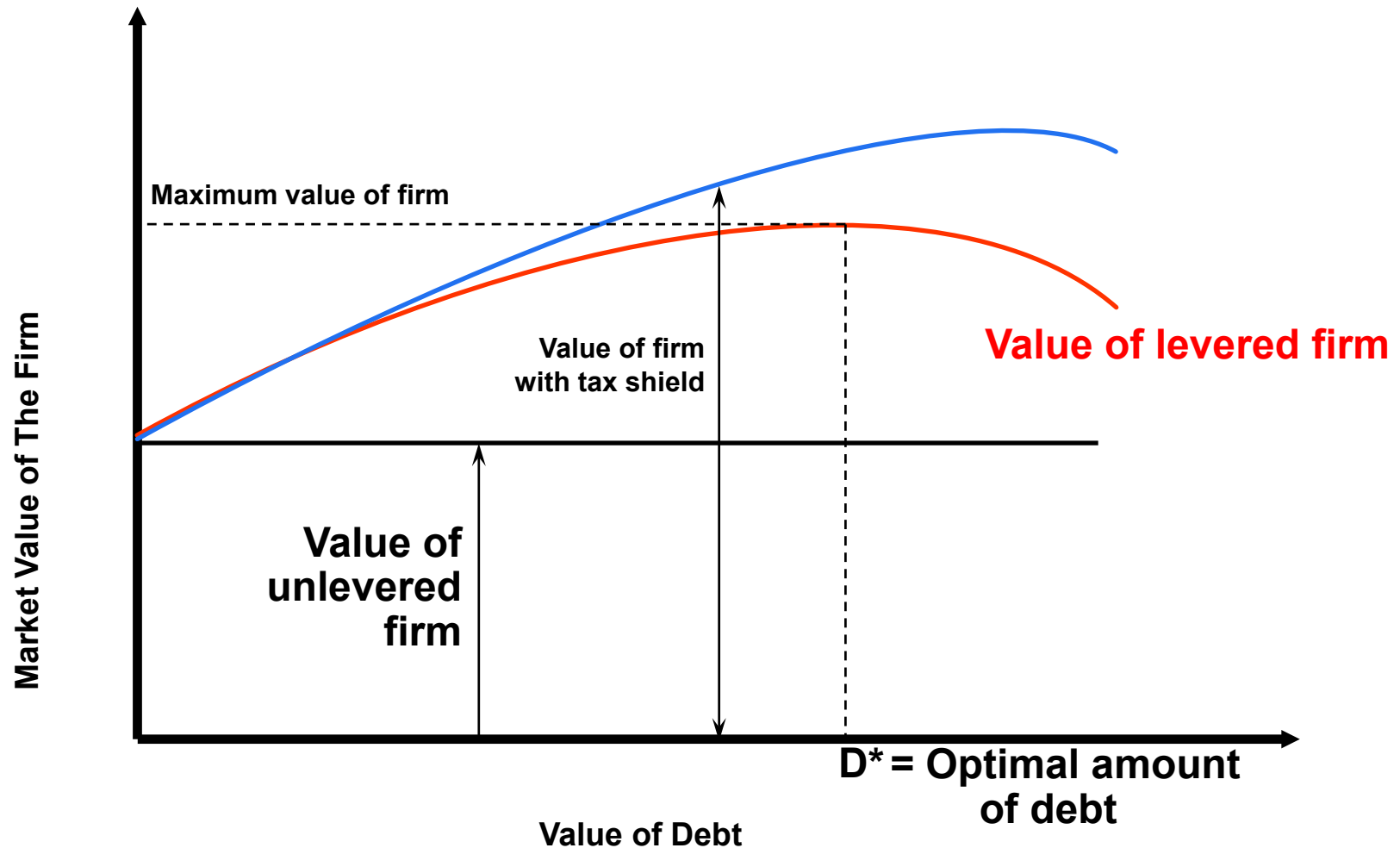
Trade-Off Theory of Capital Structure

- We can also formulate the trade-off theory in terms of the APV:

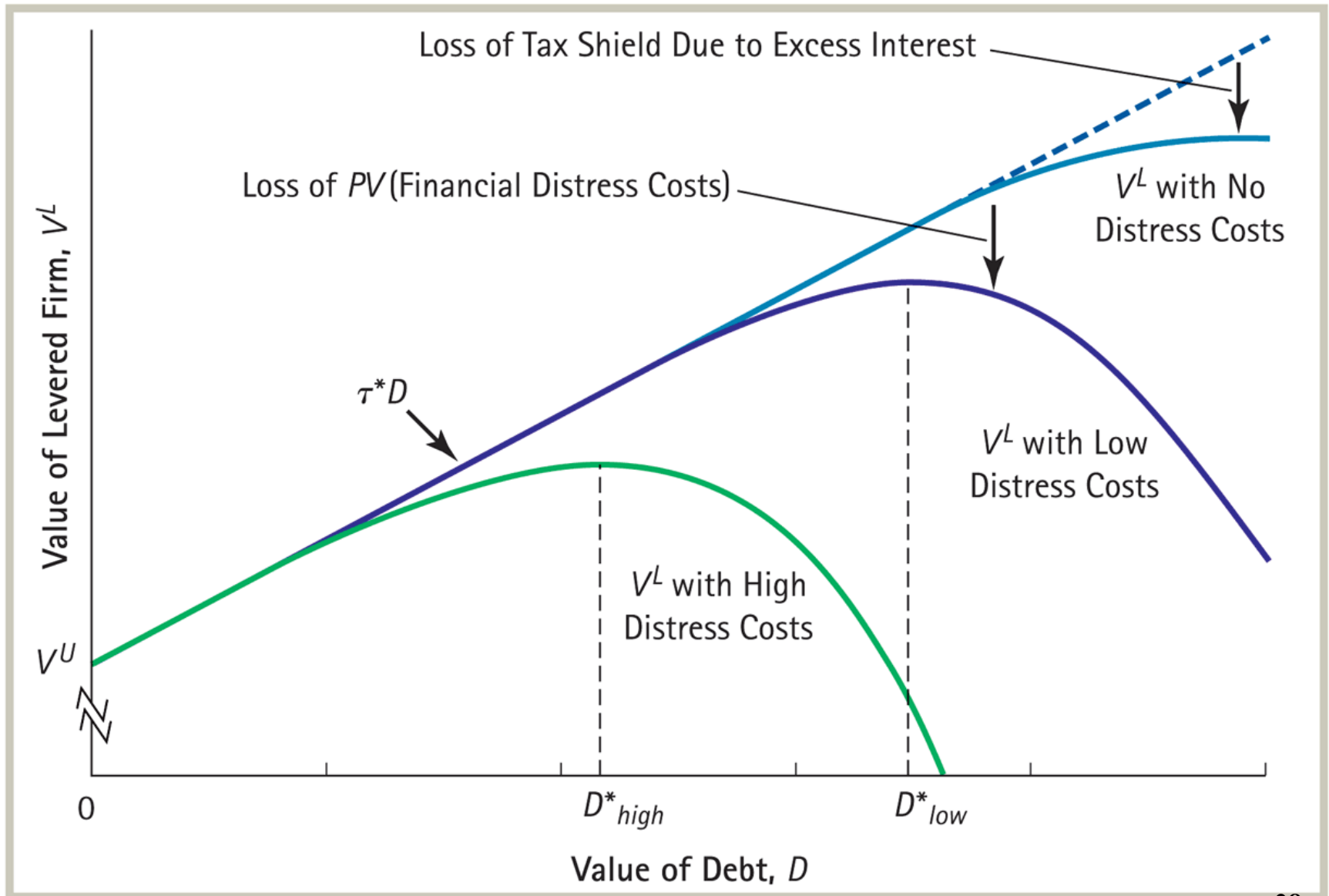
$$APV = NPV_{OA} + PV[DTS] - PV[CFD]$$

- A firm maximizes value by balancing these two financing effects
- Ex-ante, this maximizes shareholder value

Trade-Off Theory of Capital Structure



Trade-Off Theory of Capital Structure



Source: Berk and DeMarzo, Figure 16.1: p. 502 in 3e and p. 563 in 4e

Empirical Patterns in Leverage

$$y = \alpha + \beta_1 X_1 + \dots + \beta_N X_N + \varepsilon$$

Country Variable	United States	Japan	Germany	France	Italy
Panel A: Book Capital					
Tangibility	0.50*** (0.04)	1.41*** (0.18)	0.42** (0.19)	0.53** (0.26)	0.36 (0.23)
Market-to-book	-0.17*** (0.01)	-0.04 (0.04)	-0.20*** (0.07)	-0.17** (0.08)	-0.19 (0.14)
Logsale	0.06*** (0.01)	0.11*** (0.02)	-0.07*** (0.02)	0.02 (0.02)	0.02 (0.03)
Profitability	-0.41*** (0.1)	-4.26*** (0.60)	0.15 (0.52)	-0.02 (0.72)	-0.16 (0.85)
Number of observations	2079	316	175	117	96
Pseudo R ²	0.21	0.29	0.12	0.12	0.05

(Source: Rajan, Zingales (1995))

Can Trade-Off Theory Explain Patterns?

Factors that:	
<i>Increase Leverage</i>	<i>Decrease Leverage</i>
Fixed Assets Firm Size Asset Tangibility	Volatility of earnings Advertising expenditure R&D expenditure Probability of bankruptcy Uniqueness of product Investment opportunities Profitability

Takeaways: Bankruptcy is Costly

- Only economic distress matters in MM world
- Separate economic vs. financial distress when relaxing MM assumptions
 - Financial distress introduces costs (CFD)
 - CFD matters for capital structure while economic distress does not
 - Direct (Indirect) costs of CFD are small (large)
- CFD is ex-ante costly for shareholders but not for debt holders
 - The benefit of DTS and the cost of FD lead to the “Trade-Off” theory of capital structure
 - Explains broad patterns in leverage
 - Doesn’t explain why profitable firms have less leverage